

Formula Page 1: Inputs, XYZ Conversion, RGB Inversion, and Spectral Synthesis

Corresponding to Steps 1-8 in `calculate_gpu_etameric_cri.m`.

Target white point: (0.3169, 0.3310, 2.7000)

R = (0.7040, 0.2948, 0.7727)

G = (0.1705, 0.7311, 1.8770)

B = (0.1497, 0.0330, 0.2316)

W = (0.3288, 0.3376, 1.8760)

Wavelength range: 380-780 nm

Spectral sampling interval: 0.2 nm

- 1) Calibrate each channel spectrum to the specified Y value
- 2) Solve the RGB weights for the target white point using xyY-to-XYZ conversion
- 3) Synthesize the RGB target white spectrum
- 4) Normalize the W channel and RGB target white to the same luminance

$$A = [XYZ_R, XYZ_G, XYZ_B], \quad b = XYZ_{target}, \quad q = A^{-1}b$$